ADDIS ABABA SCIENCE AND TECHNOLOGY UNIVERSITY (AASTU)

Personal Profile Website Implementation Report

Laboratory Exercise Practical Technical Documentation

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1. Executive Summary

This technical report details the design, architecture, and implementation of a six-page personal portfolio website developed to fulfill the laboratory requirements for Web Technologies. The project demonstrates core proficiency in pure HTML structural standards, accessibility guidelines, media integration, interactive markup controls, and structured web navigation without external CSS frameworks or JavaScript dependency.

2. Architecture & Site Structure

The website is architected into six distinct, interconnected HTML documents linked through a persistent, uniform navigation bar. Each document addresses specific curriculum learning objectives:

File Name	Page Purpose	Demonstrated HTML Key Elements
`index.html`	Homepage & Summary	`<header>`, `<nav>`, `<main>`, `<blockquote>`, `<img>`
`about.html`	Biography & Background	`<abbr>`, `<dfn>`, `<time>`, `<dl>`, `<dt>`, `<dd>`
`skills.html`	Education & Skill Metrics	`<table>`, `<progress>`, `<meter>`, `<details>`, `<summary>`
`projects.html`	Portfolio & Downloads	`<figure>`, `<figcaption>`, nested lists, `download` links
`media.html`	Multimedia & Resources	`<audio>`, `<video>`, `<iframe>` YouTube embed

3. Key Implementation Highlights

The development strictly follows semantic markup practices, ensuring clear separation of document structure. Accessibility is enforced throughout via explicit `alt` text for image assets and explicit association between form control `id`s and `<label for="">` attributes.

4. Form Architecture & Technical Validation

The `contact.html` page implements an extensive HTML5 contact form structured logically into three `<fieldset>` components with individual `<legend>` descriptions: Personal Information, Message Details, and Optional Preferences.

> **Form Attribute & Validation Mechanics**
> The form leverages native HTML5 validation constraints without requiring JavaScript. Attributes such as `required` force complete inputs, while specific `type` values (e.g., `email`, `tel`, `url`, `date`, `color`, `range`) enforce proper input formatting natively across modern desktop and mobile browsers.

5. Directory & Asset Management

In compliance with laboratory guidelines, all external and static multimedia assets are organized systematically within dedicated subfolders to ensure maintainability and clean relative linking:

- `/` — Root folder containing all six main HTML pages (`index.html` to `contact.html`).
- `/images/` — Stores all static image assets (e.g., `profile.jpg`, project screenshots).
- `/media/` — Stores embedded multimedia files (e.g., `audio-sample.mp3`, `video-sample.mp4`).
- `/files/` — Stores downloadable documents (e.g., `sample-project-report.pdf`).

6. Verification & Compliance Checklist

The completed project was verified against all lab evaluation criteria:

1. **Global Navigation Integrity:** Verified that every page contains a functioning, identical relative menu pointing to all six site subpages.
2. **Semantic Validity:** Confirmed correct usage of structural tags (`<article>`, `<section>`, `<main>`) and document metadata.
3. **Media Playback & Fallbacks:** Verified that `<audio>`, `<video>`, and `<iframe>` render player controls and fallback message text appropriately.
4. **No Placeholder Content:** All content reflects genuine personal details, educational background, and technical ambitions.

7. Conclusion

This exercise successfully demonstrates foundational mastery of HTML5 document structure, semantic tagging, native user input handling, and clean digital resource organization, laying a solid technical groundwork for future CSS styling and client-side JavaScript integration.